

Exercise 1 : SQL Fundamentals :

SELECT & Filtering

Q1 SELECT *
FROM employees;

employees						
Id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-11-08	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2 All unique departments

SELECT DISTINCT department
FROM employees;

Department
IT
HR
Finance
Marketing

Q3 Retrieve first and last names ordered by salary descending

SELECT first-name,
last-name
FROM employees
ORDER BY salary DESC;

First-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena

Q4 Retrieve the top 3 highest-paid employees.

```

SELECT id,
       first_name,
       last_name,
       Salary
FROM employees
ORDER BY salary DESC
LIMIT 3 ;

```

id	first-name	last-name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Q5 Employees in the IT department

```

SELECT id,
       first-name,
       last-name,
       department
FROM employees
WHERE department = 'IT';

```


id	first-name	last-name	Department
5	Eva	Naidoo	IT
1	Alice	Green	IT

Question 6 : employees in Finance with salary > 60000

```

SELECT id,
       first-name,
       last-name,
       department,
FROM   employeessalary
WHERE  department = 'Finance' AND salary > 60000;

```

id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65000

Q7 employees in HR or Marketing

```

SELECT id, first-name, last-name, department
FROM   employees
WHERE  department = 'HR' OR 'Marketing';

```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8 Find employees not in IT


```

SELECT id, first-name, last-name, department
FROM employees
WHERE department NOT 'IT';

```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9 employees in IT, HR or Finance using IN

```

SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

```

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q10 SELECT id, first-name, last-name, department, salary, city
 FROM employees
 WHERE department = 'IT'
 AND salary > 65000
 AND city = 'Johannesburg';

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg